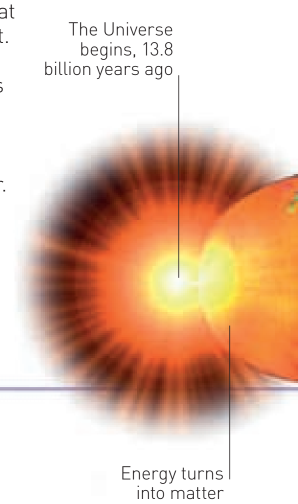
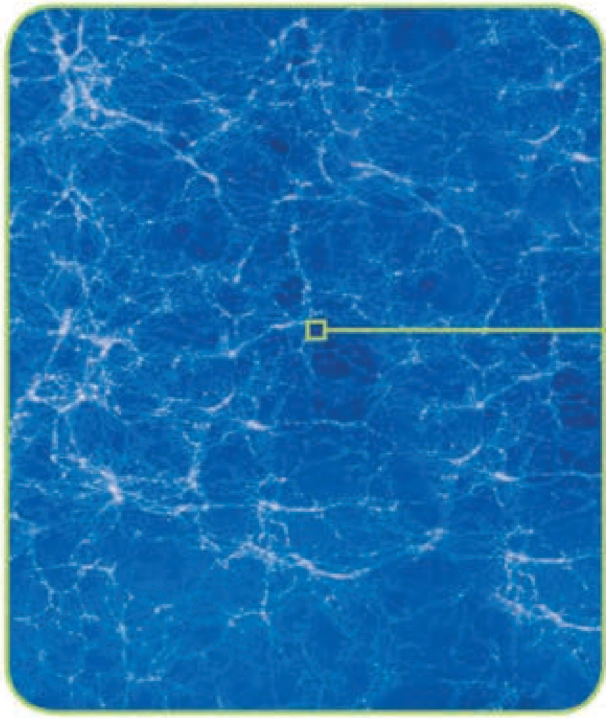
The Universe

The Universe is everything that exists – all of space, matter, energy, and time. It is a huge wide-open space with billions of galaxies, each containing billions of stars, and yet it is at least 99.99 per cent empty space. It has been expanding constantly since its beginning 13.8 billion years ago, when it exploded into life with the "Big Bang".

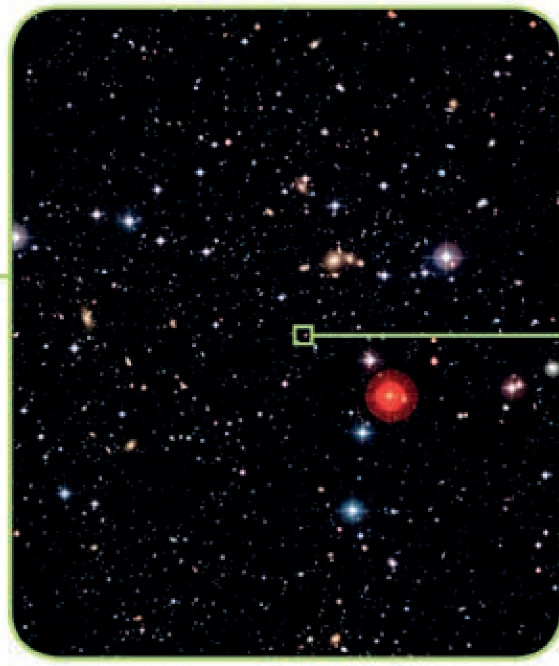
THE BIG BANG

Before the Big Bang, the entire Universe was inside a bubble that was smaller than a piece of dust. It was extremely hot and dense, and it suddenly exploded. In less than a second, the Universe became bigger than a galaxy. It carried on growing and cooling, and pure energy became matter. During the billions of years that followed, stars, planets, and galaxies formed to create the Universe as we know it.





UNIVERSE
The Universe is ever-expanding. It is full of dark energy, dark matter, and other matter such as superclusters of galaxies.



SUPERCLUSTER
Superclusters are one of the largest known structures in the Universe, made up of galaxy clusters.

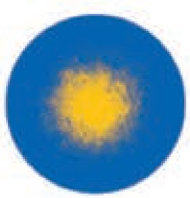
LOCAL GROUP
The Local Group is a cluster of about 50 galaxies inside the Virgo Supercluster that includes the Milky Way.

GALAXIES

Galaxies are huge groups of stars, and they can be seen in the night sky using a telescope. They come in lots of different shapes, and most of them are thought to have a massive black hole at their centre.

SPIRAL

BARRED SPIRAL



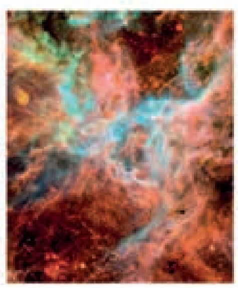
ELLIPTICAL

IRREGULAR

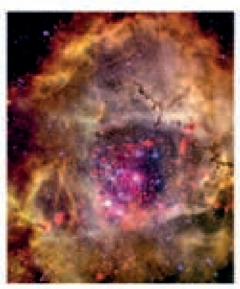
OUR GALAXY – THE MILKY WAY – IS A BARRED SPIRAL GALAXY WITH FIVE FULL OR PART ARMS

NEBULAE

Nebulae are the "nurseries" of the Universe – they are huge clouds of gas and dust in which stars form. They may be trillions of kilometres wide and many have amazing shapes and colours.



TARANTULA NEBULA



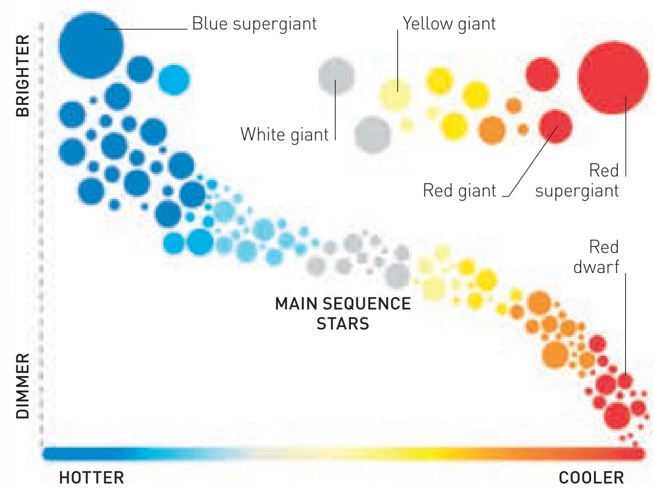
ROSETTE NEBULA

EAGLE NEBULA

N90

STARS

Stars are classified into different types depending on their temperature and brightness. Scientists use the Hertzsprung-Russell graph (shown below) to compare the size, temperature, and brightness of individual stars.



STAR TYPES
Most of the stars, including our Sun, are found along a part of the graph called the Main Sequence. As they age, these become giants or supergiants, and then dwarfs or supernovas.